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"24CS350"

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester:CIE-1/Test-1

Discrete Mathematical Structures

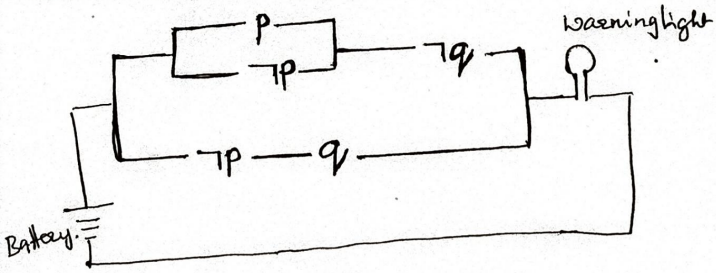
Duration:1 Hr 15 min . Date:12.11.2025 Max. Marks:30

NOTE : PART- A is Compulsory, PART- B has internal choice

PART-A

Q.NO	CO	CD	QUESTION	MARKS
1(a)	1	L3	Define Tautology and Contradiction. Check whether the compound proposition given below is Contingency, Tautology or Contradiction using truth table. $\{(p \rightarrow (q \rightarrow r)\} \rightarrow \{(p \rightarrow q) \rightarrow (p \rightarrow r)\}$	05
1(b)	1	L3	Prove the following logical equivalence using only laws of logics. $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \Leftrightarrow (p \vee q \vee r)$	05
1(c)	1	L3	Prove the following logical implication. $\{[p \wedge (p \rightarrow q) \wedge r]\} \Rightarrow \{(p \vee q) \rightarrow r\}$	05

PART-B

2(a)	1	L3	The circuit shown below includes a switching network with switches p and q, a warning light and a battery. The warning light will glow only when the switching network is closed.  i) Write the equivalent proposition for the switching network and simplify using laws of logics. ii) Construct the truth table for the simplified expression and determine the condition(s) of switch p and q under which the warning light glow. iii) Draw the simplified switching network.	08
2(b)	1	L3	Establish the validity of the following argument. $[(p \rightarrow q) \wedge (q \rightarrow (r \wedge s)) \wedge (\neg r \vee (\neg t \vee u)) \wedge (p \wedge t)] \Rightarrow u$	07

			OR	
3(a)	**1**	**L3**	Establish the validity of the following arguments using Inference rules. 1. I will buy a new phone or I will buy a new laptop. 2. If I buy a laptop, I will not buy a tablet. 3. I did not buy a phone. Conclusion: Therefore, I didn't buy a tablet. Also prove the same using Truth Table.	**07**
3(b)	**1**	**L3**	i) Prove the following logical equivalence using laws of logics. $(p \downarrow q) \Leftrightarrow (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ ii) Prove that NAND and NOR Connectives are not associative.	**4+4**

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Use the knowledge needed to test the logic of the program.	L3
CO2	Demonstrate the concept of relations and be able to determine their properties.	L3
CO3	Explain and apply decode group codes based on the various methods.	L3
CO4	Identify different types of graphs such as Isomorphism, Bipartite, planar, Hamiltonian paths and cycles, Euler trails and circuits based on their properties.	L3
CO5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.	L3